

Production Rules of FSA

Friday, October 31, 2025

8:18 AM

Grammar (G) can be described with 4 tuples:

- V: Variables
- T: Terminal Symbol
- P: Production Rules for V and T
- S: Start Symbols

Chomsky hierarchy:

We Will Only learn Type 2 & 3.

Right-Linear

The non-terminal
is placed on the
Right
 $A \rightarrow xB$

Left-linear

On the left

$A \rightarrow Bx$

I.e. of a Production Rule

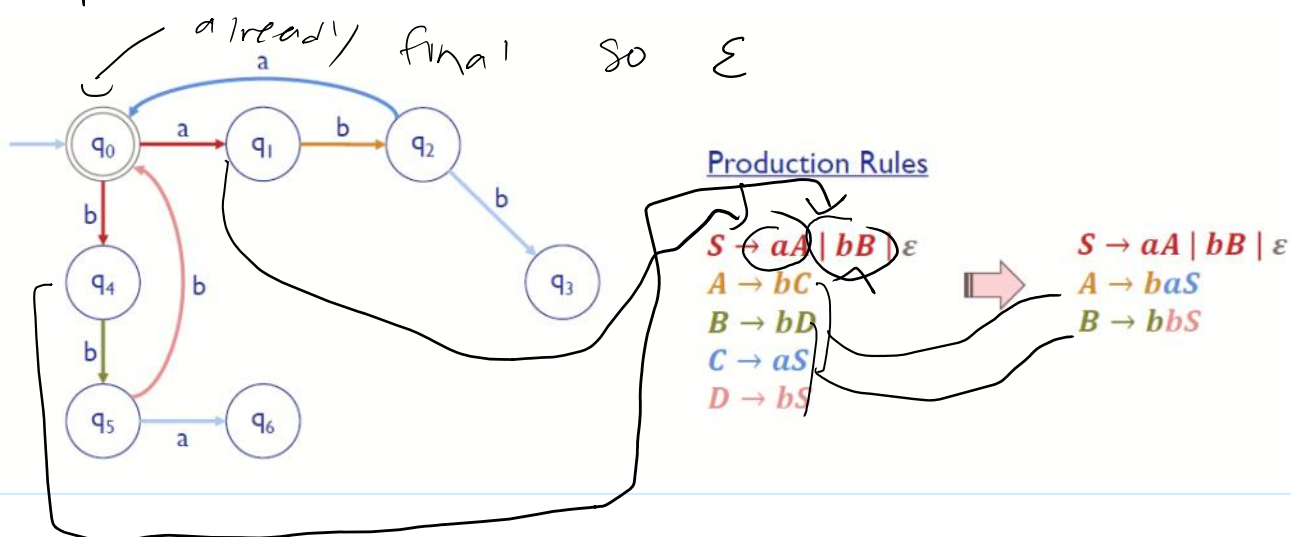
$S \rightarrow aA$

$A \rightarrow B | C$

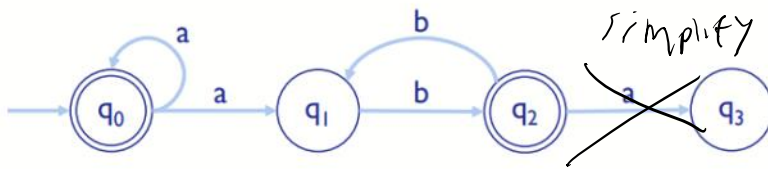
$B \rightarrow aB | b$

from start, input a, go to
production Rule A

Simplification



1. Construct the regular grammar corresponding to the following FSA.

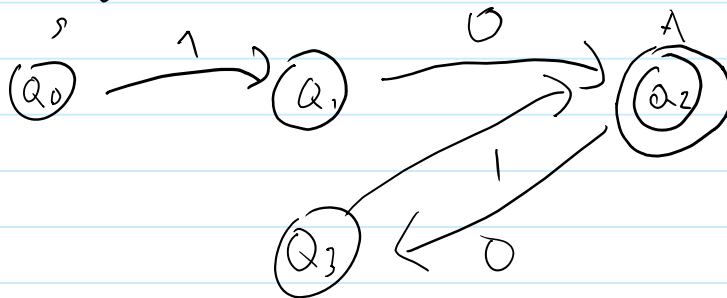


$V = a, b$
 $T = A, B, C$
 $P =$

$S \rightarrow \epsilon a aA$	$S \rightarrow \epsilon a aA$
$A \rightarrow b bB$	$A \rightarrow b bB$
$B \rightarrow \epsilon bA aC$	$B \rightarrow \epsilon bA$

Edge case

$S \rightarrow 10A$
 $A \rightarrow 01A | \epsilon$



telita

CFG Simplification

Friday, November 14, 2025

8:38 AM

1. Eliminate useless productions from the CFG below.

$S \rightarrow \cancel{AB} \mid CA$

$A \rightarrow a$

$\cancel{B \rightarrow BC \mid AB}$

$C \rightarrow \cancel{aB} \mid b$

2. Eliminate useless productions from the CFG below.

$S \rightarrow aS \mid A \mid \cancel{\epsilon}$

$A \rightarrow a$

$\cancel{B \rightarrow aa}$

$\cancel{C \rightarrow aCb}$

$S \rightarrow aS \mid a$

How to Simplify:

- find Useless productions (non-terminals that aren't called)
- Eliminate Unit Productions that only has 1 single terminal
- Eliminate Null(ϵ) productions

$S \rightarrow Ab \mid Cd$
 $A \rightarrow d$
 $C \rightarrow \epsilon$

$\Rightarrow S \rightarrow Ab \mid d$
 $A \rightarrow d$

$\rightarrow$ Btm Ind (optional)
 $S \rightarrow db \mid d$

What's Considered Useless?

- Variables / terminals that doesn't appear anywhere
- Non-Generating
- Non-Reachable Variables

Simplification of CFG:

- Elimination of Useless productions

What's considered useless?

$\hookrightarrow$ Variables that doesn't appear anywhere

$\hookrightarrow$ Non-Reachable Variables

$\hookrightarrow$ Non-Generating Symbols

$S \rightarrow aA$

$\cancel{C \rightarrow Bc}$

$S \rightarrow aA$

$A \rightarrow c \mid \cancel{B}$

A symbol becomes non-generating when **every** one of its productions keeps at least one nonterminal

that also cannot terminate.

I.e. $\rightarrow$ the E is never gone

$E \rightarrow aEb$

OR

$\rightarrow$ the B is never gone

$B \rightarrow BC \mid AB$

- Elimination of Unit Productions

Eliminate symbols that only contain terminals, combine them with their references

CFG

$S \rightarrow Sb \mid C$

$C \rightarrow D \mid ef$

$D \rightarrow dd$

After Elimination

$S \rightarrow Sb \mid dd \mid ef$

$C \rightarrow dd \mid ef$

~~$D \rightarrow dd$~~

$S \rightarrow aSaA \mid A$

$A \rightarrow abA \mid b$



$S \rightarrow aSaA \mid abA \mid b$

$A \rightarrow abA \mid b$

$S \rightarrow ABC$

$C \rightarrow \epsilon \mid A$

$B \rightarrow \epsilon \mid S$

$S \rightarrow ASA \mid AA \mid AS \mid A$

$ABC \mid A\epsilon C \mid AB\epsilon \mid A\epsilon\epsilon$

$ASA \mid AA \mid AS \mid A$

CNF, and CFG conversion to CNF

Friday, November 21, 2025

8:40 AM

$S \rightarrow bA | aB$
 $A \rightarrow bAA | aS | a$
 $B \rightarrow aBB | bS | b$

CNF harus ada antara :
 - 2 Variables atau
 - 1 terminal

$P_1 \rightarrow b$ $P_3 \rightarrow AA$
 $P_2 \rightarrow a$ $P_4 \rightarrow bB$

$S \rightarrow P_1 A | P_2 B$
 $A \rightarrow P_1 AA | P_2 S | a$
 $B \rightarrow P_2 bB | P_1 S | b$

CFG

- $S \rightarrow aAB | ch | CD$
- $A \rightarrow dbE | eEC$
- $B \rightarrow ff | DD$
- $C \rightarrow ADB | aS$
- $D \rightarrow i$
- $E \rightarrow jD$

$P_1 = a$ $P_2 = c$ $S \rightarrow P_1 P_8 | P_2 P_{11} | CD$
 $P_3 = ch$ $P_4 = h$ $A \rightarrow P_3 P_{12} E | P_4 P_9$ $\rightarrow P_{13} = P_{12} E$
 $P_5 = db$ $P_6 = b$ $B \rightarrow P_5 P_{10} | DD$
 $P_7 = e$
 $P_8 = DB$ $C \rightarrow A P_5 | R S$ $P_3 P_{13}$
 $P_9 = a$ $D \rightarrow i$
 $P_{10} = j$ $E \rightarrow P_7 D$
 $P_{11} = AB$
 $P_{12} = EC$
 $P_{13} = f$
 ugh benar kecuali number prod. nya

CFG

- $S \rightarrow aAB | ch | CD$
- $A \rightarrow dbE | eEC$
- $B \rightarrow ff | DD$
- $C \rightarrow ADB | aS$
- $D \rightarrow i$
- $E \rightarrow jD$

$P_1 = a$ $P_8 = e$
 $P_2 = AB$ $P_9 = EC$
 $P_3 = c$ $P_{10} = f$
 $P_4 = h$ $P_{11} = DB$
 $P_5 = d$ $P_{12} = j$
 $P_6 = P_7 E$
 $P_7 = b$

$S \rightarrow P_1 P_2 | P_3 P_4 | CD$
 $A \rightarrow P_5 P_6 | P_8 P_9$
 $B \rightarrow P_{10} P_{11} | DD$
 $C \rightarrow A P_{11} | P_1 S$
 $D \rightarrow i$
 $E \rightarrow jD$

$C \rightarrow H H \mid I \mid J$

$D \rightarrow I$

$E \rightarrow P_1 D$

Exercise

1. Put the CFG below into CNF.

$S \rightarrow aSb \mid ab$

$P_1 = a$

$P_2 = S P_3$

$P_3 = b$

$S \rightarrow P_1 P_2 \mid P_1 P_3$

2. Put the CFG below into CNF.

$S \rightarrow aSaA \mid A$

$A \rightarrow abA \mid b$

$S \rightarrow aSaA \mid abA \mid b$
 $A \rightarrow abA \mid b$

$P_1 = a$

$P_2 = S P_1$

$P_3 = P_2 A$

$P_4 = P_3 A$

$P_5 = b$

$S \rightarrow P_1 P_2 \mid P_4 A \mid b$
 $A \rightarrow P_4 A \mid b$

3. Put the CFG below into CNF.

$S \rightarrow abAB$

$A \rightarrow bAB \mid \epsilon$

$B \rightarrow BAa \mid A \mid \epsilon$

$S \rightarrow abAB \mid abAA \mid abA \mid ab$
 $A \rightarrow bAB \mid bA \mid bB \mid b$
 $B \rightarrow BAa \mid A$

$S \rightarrow abAB \mid abAA \mid abA \mid ab$
 $A \rightarrow bAB \mid bA \mid bB \mid b$
 $B \rightarrow BAa \mid bAB \mid bA \mid bB \mid b \mid Ba \mid Aa \mid a$

$P_1 = a$

$P_2 = P_3 P_4$

$P_3 = b$

$P_4 = AB$

$P_5 = P_3 P_6$

$P_6 = AA$

$P_7 = P_3 A$

$P_8 = BA$

$S \rightarrow P_1 P_2 \mid P_1 P_5 \mid P_1 P_7 \mid P_1 P_3$
 $A \rightarrow P_3 P_4 \mid P_3 A \mid P_3 B \mid b$
 $B \rightarrow P_8 P_1 \mid P_3 P_4 \mid P_3 A \mid P_3 B \mid P_3 P_1 \mid A P_1 \mid a \mid b$

Elimination of left Recursion

Friday, November 28, 2025 8:29 AM

live recursion
 $S \rightarrow \underline{Sd} \mid \underline{a}$
 $\alpha \quad \beta$ $\hookrightarrow$ Hasi'

$S \rightarrow a \mid aZ$
 $Z \rightarrow d \mid dZ$

live recursion

$S \rightarrow \underline{Sab} \mid \underline{aSc} \mid \underline{dd} \mid \underline{ff} \mid \underline{Sbd}$
 $\alpha_1 \quad \beta_1 \quad \beta_2 \quad \beta_3 \quad \alpha_2$

$S \rightarrow \underline{aSc} \mid \underline{aScZ} \mid \underline{ddZ} \mid \underline{ffZ} \mid \underline{dd} \mid \underline{ff}$
 $1 \quad 2 \quad 3 \quad 2 \quad 3$

$Z \rightarrow ab \mid bd \mid abZ \mid bdZ$

TO CFG
 $S \rightarrow \underline{Sa} \mid \underline{aAc} \mid \underline{c} \mid \underline{\epsilon}$
 $A \rightarrow \underline{Ab} \mid \underline{ba}$
 $\alpha_1 \quad \beta_1 \quad \beta_2 \quad \beta_3$

$S \rightarrow aAc \mid c \mid \epsilon \mid aAcZ_1 \mid cZ_1 \mid Z_1$
 $Z_1 \rightarrow a \mid aZ_1$
 $A \rightarrow ba \mid baZ_2$
 $Z_2 \rightarrow b \mid bZ_2$

Practice

1. $A \rightarrow \underline{Aa} \mid \underline{aBc}$
 $\alpha_1 \quad \beta_1$

$$A \rightarrow aBc \mid aBcZ_1$$

$$Z_1 \rightarrow Aa \mid AaZ_1$$

2.

$$\begin{array}{l}
 A \rightarrow A \overset{\alpha_1}{b} \overset{\beta_1}{a} B \mid \overset{\beta_1}{\epsilon} \\
 B \rightarrow B \overset{\alpha_1}{A} \overset{\beta_1}{a} \mid \overset{\beta_2}{A} \mid \overset{\beta_2}{\epsilon}
 \end{array}$$

m

$$A \rightarrow Z_1 \mid \epsilon$$

$$Z_1 \rightarrow baB \mid baB$$

$$\begin{array}{l}
 B \rightarrow A \mid \epsilon \mid AZ_2 \mid Z_2 \\
 Z_2 \rightarrow Aa \mid AaZ_2
 \end{array}$$

CNF to GNF

Friday, December 5, 2025

8:39 AM

$S \rightarrow CA$ — Was allowed but since it uses C, substitute it
 $A \rightarrow a | d$
 $B \rightarrow b$
 $C \rightarrow DD$ — Was allowed but since it uses D, which was dirty, substitute it.
 $D \rightarrow AB$ — A is smaller than D, substitute value

If it's CFG, Convert to CNF first!

Step 1: Order it where $S < A < B < C < D < \dots$
 Work from the bottom up!

$D \rightarrow aB | dB$

B is smaller than D yes
 but it starts with a terminal
 So it's alright

Substitute the first D

$C \rightarrow DD$

$C \rightarrow aBD | dBD$

Substitute C

$S \rightarrow CA$

$S \rightarrow aBDA | dBDA$

$S \rightarrow aBPA | dBDA$

$A \rightarrow a | d$

$B \rightarrow b$

$C \rightarrow aBD | dBD$

$D \rightarrow aB | dB$

} final Answer

Dealing with left recursion

$A \rightarrow BC$

$B \rightarrow CA | b$ — Substitute C since C was cleaned $\rightarrow B \rightarrow bCBA | aA | bCBZA | aZA | b$

$C \rightarrow AB | a$ $C \neq A$

Substitute A

Substitute B

Substitute A $\swarrow$

$$C \rightarrow BCB \mid a$$

Substitute B

$$C \rightarrow \underline{CACB} \mid \underline{bCB} \mid \underline{a}$$

α_1

β_1

β_2

$$C \rightarrow bCB \mid a \mid bCBZ \mid aZ$$

$$Z \rightarrow ACB \mid ACBZ$$

(Step 3)

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Substitute B since it was cleaned

$$A \rightarrow bCBAC \mid aAC \mid bCBZAC \mid aZAC \mid b$$

$$A \rightarrow bCBAC \mid aAC \mid bCBZAC \mid aZAC \mid b$$

$$B \rightarrow bCBA \mid aA \mid bCBZA \mid aZA \mid b$$

$$C \rightarrow bCB \mid a \mid bCBZ \mid aZ$$

$$Z \rightarrow ACB \mid ACBZ$$

} final answer

$$\delta(q, a, X) = (p, \gamma)$$

This means:

If the PDA is in state q , reads input symbol a (or ϵ), and sees stack top X , then it can move to state p and replace X with the string γ on the stack.

ACCEPTANCE BY FINAL STATE

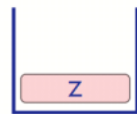
$P = (\{q_1, q_2\}, \{a, b\}, \{A, B, Z\}, \delta, q_1, Z, \{q_2\})$

String *abba*:

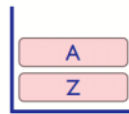
Transition function δ :

$\delta(q_1, \epsilon, Z) = (q_2, Z)$
 $\delta(q_1, a, Z) = (q_1, AZ)$
 $\delta(q_1, b, Z) = (q_1, BZ)$
 $\delta(q_1, a, A) = (q_1, AA)$
 $\delta(q_1, b, A) = (q_1, \epsilon)$
 $\delta(q_1, a, B) = (q_1, \epsilon)$
 $\delta(q_1, b, B) = (q_1, BB)$

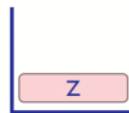
Stack:



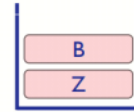
1. $\delta(q_1, a, Z) = (q_1, AZ)$



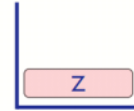
2. $\delta(q_1, b, A) = (q_1, \epsilon)$



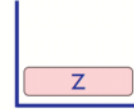
3. $\delta(q_1, b, Z) = (q_1, BZ)$



4. $\delta(q_1, a, B) = (q_1, \epsilon)$



5. $\delta(q_1, \epsilon, Z) = (q_2, Z)$



14 PUSHDOWN AUTOMATA (PDA)

ACCEPTANCE BY EMPTY STACK

$P = (\{q_1, q_2\}, \{0, 1, 2\}, \{Z, B, G\}, \delta, q_1, Z, \emptyset)$

String 020:

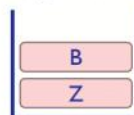
Transition function δ :

$\delta(q_1, 0, Z) = (q_1, BZ)$
 $\delta(q_1, 0, B) = (q_1, BB)$
 $\delta(q_1, 0, G) = (q_1, BG)$
 $\delta(q_1, 2, Z) = (q_2, Z)$
 $\delta(q_1, 2, B) = (q_2, B)$
 $\delta(q_1, 2, G) = (q_2, G)$
 $\delta(q_2, 0, B) = (q_2, \epsilon)$
 $\delta(q_2, \epsilon, Z) = (q_2, \epsilon)$
 $\delta(q_1, 1, Z) = (q_1, GZ)$
 $\delta(q_1, 1, B) = (q_1, GB)$
 $\delta(q_1, 1, G) = (q_1, GG)$
 $\delta(q_2, 1, G) = (q_2, \epsilon)$

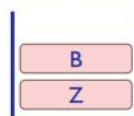
Stack:



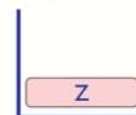
1. $\delta(q_1, 0, Z) = (q_1, BZ)$



2. $\delta(q_1, 2, B) = (q_2, B)$



3. $\delta(q_2, 0, B) = (q_2, \epsilon)$



4. $\delta(q_2, \epsilon, Z) = (q_2, \epsilon)$

